

# Dongfang Zhao, Ph.D.

## Professional Preparation

|                                               |                             |      |
|-----------------------------------------------|-----------------------------|------|
| Katholieke Universiteit Leuven, Belgium       | Statistics, Cum Laude       | 2005 |
| Emory University, Atlanta, GA                 | Computer Science, Master    | 2008 |
| Illinois Institute of Technology, Chicago, IL | Computer Science, Doctorate | 2015 |
| University of Washington, Seattle, WA         | Computer Science, PostDoc   | 2017 |

## Employment

|              |                                                                                    |
|--------------|------------------------------------------------------------------------------------|
| 2018–present | Assistant Professor (tenure-track) of Computer Science, University of Nevada, Reno |
| 2015         | Research Intern, IBM Almaden Research Center, San Jose, CA                         |
| 2014         | Research Intern, Argonne National Laboratory, Chicago, IL                          |
| 2013         | Research Intern, Pacific Northwest National Laboratory, Seattle, WA                |
| 2009–2011    | Software Developer (full-time), Epic Systems Corporation, Madison, WI              |

## Awards and Grants

|      |                                                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | Federal Research Grant, NSF 2112345, \$255,916, PI at University of Nevada (50% share)<br>“A DLT Machine Learning Platform for Blockchain Warehousing”                   |
| 2020 | Federal Research Grant, DOE SC0020455, \$200,000, PI at University of Nevada (50% share)<br>“HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”             |
| 2019 | Best Paper Award, International Conference on Cloud Computing<br>“BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”                 |
| 2019 | Industrial Research Grant, Amazon Web Services, \$30,000                                                                                                                 |
| 2019 | Industrial Research Grant, Google Cloud Platform, \$15,000                                                                                                               |
| 2018 | Best Student Paper Award, IEEE International Conference on Cloud Computing<br>“Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory” |
| 2017 | Industrial Research Grant, Microsoft Azure, \$20,000                                                                                                                     |
| 2017 | Industrial Research Grant, Amazon Web Services, \$10,000                                                                                                                 |
| 2016 | Postdoctoral Fellowship, Sloan Foundation, \$155,000                                                                                                                     |

## Services

|              |                      |                                                     |
|--------------|----------------------|-----------------------------------------------------|
| 2021–present | Review Board         | IEEE Transactions on Distributed & Parallel Systems |
| 2022         | Program Committee    | AAAI Conference on Artificial Intelligence          |
| 2022         | Grant Review Panel   | National Science Foundation: CISE OAC               |
| 2022         | Grant Review Panel   | National Science Foundation: CISE IIS               |
| 2019         | Organizing Committee | ACM/IFIP International Middleware Conference        |
| 2018         | Grant Review Panel   | U.S. Department of Energy: ASCR                     |
| 2018         | Grant Review Panel   | National Science Foundation: CISE IIS               |

## Teaching

|                                     |                            |              |
|-------------------------------------|----------------------------|--------------|
| CS 705: Cryptography and Blockchain | University of Nevada, Reno | 2022–present |
| CS 791: Distributed Computing       | University of Nevada, Reno | 2020–present |
| CS 457: Database Management Systems | University of Nevada, Reno | 2018–present |

## Selected Publications

(**Dongfang Zhao** is the student's \*PhD advisor, <sup>†</sup>thesis committee member, or <sup>‡</sup>postdoc collaborator)

- [1] \*Olamide Timothy Tawose, Jun Dai, Lei Yang, **Dongfang Zhao**. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. ACM International Conference on Management of Data (SIGMOD), 2023.
- [2] \*Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**: Memory Scaling of Cloud-based Big Data Systems: A Hybrid Approach. IEEE Transactions on Big Data. 8(5): 1259-1272 (2022).
- [3] \*Abdullah Al-Mamun, Haoting Shen, **Dongfang Zhao**. DEAN: A Lightweight and Resource-efficient Blockchain Protocol for Reliable Edge Computing. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.
- [4] \*Olamide Timothy Tawose, \*Bin Li, Lei Yang, Feng Yan, **Dongfang Zhao**. Topological Modeling and Parallelization of Multidimensional Data on Microelectrode Arrays. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022. Direct acceptance without revision
- [5] \*Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [6] \*Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021. Direct acceptance without revision
- [7] \*Xinying Wang, \*Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**: HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020. Oral presentation
- [8] **Dongfang Zhao**, Mohamed Mohamed, Heiko Ludwig: Locality-Aware Scheduling for Containers in Cloud Computing. IEEE Transactions on Cloud Computing. 8(2): 635-646 (2020).
- [9] \*Xinying Wang, \*Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud. The 12th International Conference on Cloud Computing (CLOUD), 2019. Best paper award
- [10] <sup>†</sup>Heyang Qin, <sup>†</sup>Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan: Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [11] \*Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**: Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory. The 11th IEEE International Conference on Cloud Computing (IEEE CLOUD), 2018. Best student paper award

- [12] \*Jiayao Wang, \*Abdullah Al-Mamun, Tonglin Li, Linhua Jiang, **Dongfang Zhao**: Toward Performant and Energy-efficient Queries in Three-tier Wireless Sensor Networks. Proceedings of the 47th International Conference on Parallel Processing (ICPP), 2018.
- [13] ‡Parmita Mehta, ‡Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad: Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. Proceedings of the VLDB Endowment (VLDB). 10(11): 1226-1237 (2017).
- [14] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Zhihan Lu, Xiaohua Xu: Toward Efficient and Flexible Metadata Indexing of Big Data Systems. IEEE Transactions on Big Data. 3(1): 107-117 (2017).
- [15] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Xiaobing Zhou, Ioan Raicu: Exploiting multi-cores for efficient interchange of large messages in distributed systems. Concurrency and Computation: Practice and Experience. 28(13): 3568-3585 (2016).
- [16] **Dongfang Zhao**, Ke Wang, Kan Qiao, Tonglin Li, Iman Sadooghi, Ioan Raicu: Toward high-performance key-value stores through GPU encoding and locality-aware encoding. Journal of Parallel and Distributed Computing. 96: 27-37 (2016).
- [17] Zhou Zhou, Xu Yang, **Dongfang Zhao**, Paul Rich, Wei Tang, Jia Wang, Zhiling Lan: I/O-aware bandwidth allocation for petascale computing systems. Parallel Computing. 58: 107-116 (2016).
- [18] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu: Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems. 27(6): 1824-1837 (2016).
- [19] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu: Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. IEEE Transactions on Service Computing 9(1): 96-109 (2016).
- [20] Zhou Zhou, Xu Yang, **Dongfang Zhao**, Paul Rich, Wei Tang, Jia Wang, Zhiling Lan: I/O-Aware Batch Scheduling for Petascale Computing Systems. IEEE International Conference on Cluster Computing (CLUSTER), 2015.
- [21] Tonglin Li, Xiaobing Zhou, Kevin Brandstatter, **Dongfang Zhao**, Ke Wang, Anupam Rajendran, Zhao Zhang, Ioan Raicu: ZHT: A Light-Weight Reliable Persistent Dynamic Scalable Zero-Hop Distributed Hash Table. The 27th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2013.
- [22] **Dongfang Zhao**, Kent Burlingame, Corentin Debains, Pedro Alvarez-Tabio, Ioan Raicu: Towards high-performance and cost-effective distributed storage systems with information dispersal algorithms. IEEE International Conference on Cluster Computing (CLUSTER), 2013.
- [23] **Dongfang Zhao**, Chen Shou, Tanu Malik, Ioan Raicu: Distributed data provenance for large-scale data-intensive computing. IEEE International Conference on Cluster Computing (CLUSTER), 2013.
- [24] **Dongfang Zhao**, Li Yang: Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence. 31(1): 86-98 (2009).